P1) $A(2,-3,0)$ $\quad \vec{v} = \vec{i} - \vec{j} + \vec{k}$

$P: \; a(x-x_A) + b(y-y_A) + c(z-z_A) = 0$

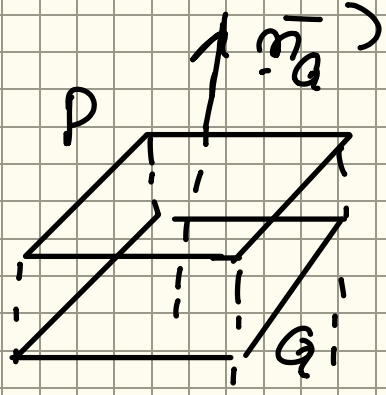
$\Leftrightarrow \; x-2-y-3+z-0=0$

$x-y+z-5=0$

P2) $M(1,-1,2)$

$Q: 2x+y-4z+5=0$

$\vec{n}_Q = 2\vec{i} + \vec{j} - 4\vec{k}$



$Q \parallel P \Rightarrow \vec{n} \perp P \Rightarrow$ $P: 2(x-1)+1(y+1)-4(z-2)=0$

$\vec{n} \perp Q$ $\quad 2x-2+y+1-4z+8=0$

$2x+y-4z+7=0$

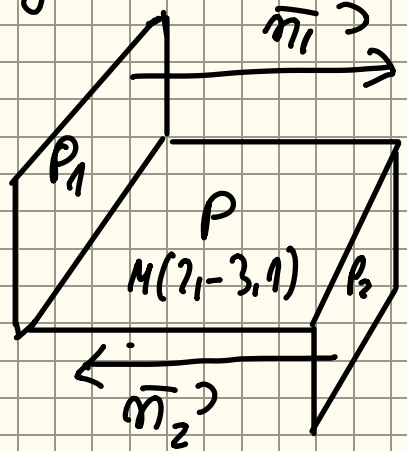
P4) $M(2,-3,1)$

$P_1: x-y+2z-5=0$

$P_2: 2x-3y-z+4=0$

$\vec{n}_1 = \vec{i} - \vec{j} + 2\vec{k}$

$\vec{n}_2 = 2\vec{i} - 3\vec{j} - \vec{k}$



$P \perp P_1$, $\vec{n}_1 \perp P_1 \Rightarrow \vec{n}_1 \parallel P$

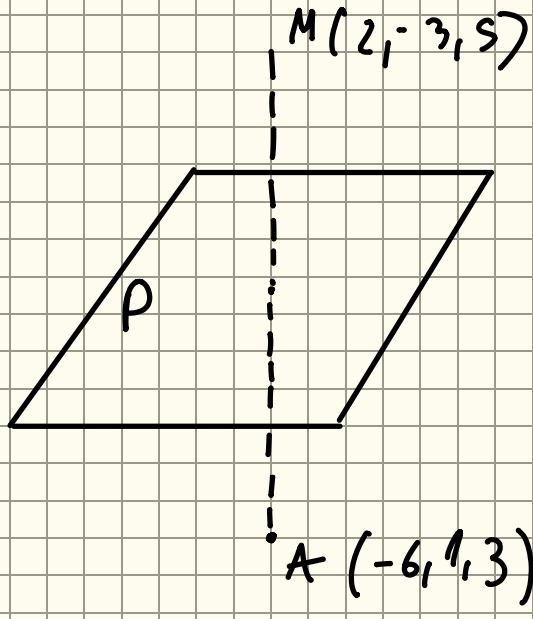
$P \perp P_2$, $\vec{n}_2 \perp P_2 \Rightarrow \vec{n}_2 \parallel P$

$M \in P$

$$\Rightarrow \begin{vmatrix} x-2 & y+3 & z-1 \\ 1 & -1 & 2 \\ 2 & -3 & -1 \end{vmatrix} = 0$$

$\Leftrightarrow \quad 7x - 14 + 5y + 15 - z + 1 = 0$

$\Leftrightarrow \quad 7x + 5y - z + 2 = 0$



P5) $M(2, -3, 5)$

$A(-6, 1, 3)$

$A \cdot M$ nym $\Rightarrow \vec{AM} \perp P$

$\Rightarrow \vec{AM} = 8\vec{i} - 4\vec{j} + 2\vec{k}$

$N(-2, -1, 4) \Rightarrow P: 8(x+2) - 4(y+1) + 2(z-4) = 0$

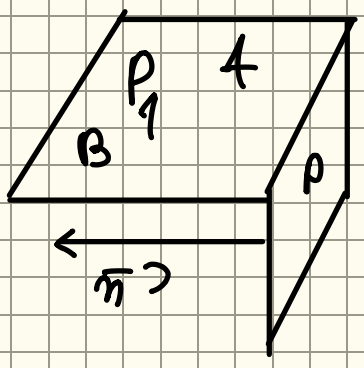
$\Leftrightarrow \quad 8x + 16 - 4y - 4 + 2z - 8 = 0$

$\Leftrightarrow \quad 8x - 4y + 2z + 4 = 0 \quad /:2$

$\Leftrightarrow \quad 4x - 2y + z + 2 = 0$

P6) $A(2, -1, 3)$ a) $P: x - y + 3z + 5 = 0$

$B(1, -3, 5)$



$\vec{n} = \vec{i} - \vec{j} + 3\vec{k}$

$\left.\begin{array}{l} \vec{n} \perp P \\ P \perp P_1 \end{array}\right| \Rightarrow \vec{n} \parallel P$

$\vec{AB} = -\vec{i} - 2\vec{j} + 2\vec{k}$

$\vec{n_{P_1}} = \vec{AB} \times \vec{n} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -1 & -2 & 2 \\ 1 & -1 & 3 \end{vmatrix} = -8\vec{i} + 5\vec{j} + 3\vec{k}$

$\Rightarrow P: -4(x-2)+5(y+1)+3(z-3)=0$

$\Leftrightarrow -4x+8+5y+5+3z-9=0$

$\Leftrightarrow -4x+5y+3z+4=0$

b) $\vec{v}=2\vec{i}-3\vec{j}+4\vec{k}$

$P \parallel \vec{v} \Rightarrow \vec{AB} \times \vec{v} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -1 & -2 & 2 \\ 2 & -3 & 4 \end{vmatrix} = -2\vec{i}+8\vec{j}+7\vec{k}$

$\Rightarrow -2(x-2)+8(y+1)+7(z-3)=0$

$-2x+4+8y+8+7z-21=0$

$-2x+8y+7x-9=0$

L2) $M(2,-1,5)$

$\vec{v}=3\vec{i}-5\vec{j}+\vec{k}$

$D: \dfrac{x-2}{3}=\dfrac{x+1}{-5}=\dfrac{z-5}{1}=t, t\in\mathbb{R}$

L3) $[AB]$ $A(2,-1,3)$

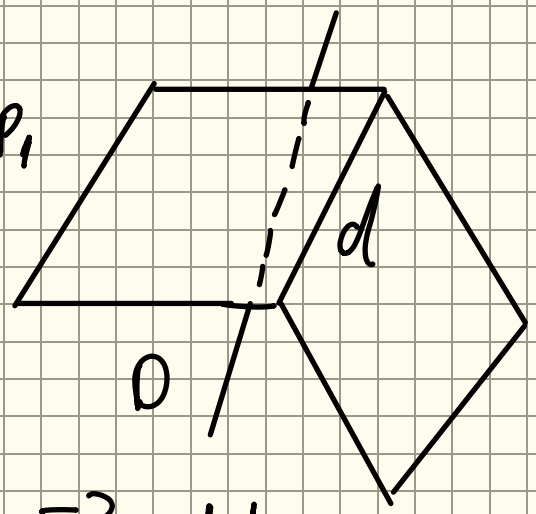
$B(-2,1,5)$

$\Rightarrow \dfrac{x-x_A}{x_B-x_A}=\dfrac{y-y_A}{y_B-y_A}=\dfrac{z-z_A}{z_B-z_A}=t$

$\dfrac{x-2}{-4}=\dfrac{y+1}{2}=\dfrac{z-3}{2}=t$

$$\Rightarrow \begin{cases} x = 2-4t \\ y = -1+2t \\ z = 3+2t \end{cases} \quad \begin{matrix} t=0 \Rightarrow A \\ t=1 \Rightarrow B \end{matrix} \Big| \Rightarrow 0 \le t \le 1$$

L4) $M(2,-3,4)$, $d: \begin{cases} x-y+z-3=0 : P_1 \\ x+y-4=0 : P_2 \end{cases}$



$d = P_1 \cap P_2$

$D \parallel d$

$\vec{n_1} = \vec{i} - \vec{j} + \vec{k}$

$\vec{n_2} = \vec{i} + \vec{j}$

$\vec{n_1} \perp d$

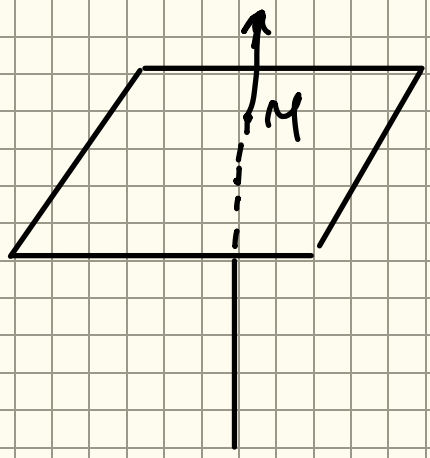
$\vec{n_2} \perp d \Rightarrow$ d || simultan pe $\vec{n_1}$ și $\vec{n_2}$

$$\vec{n_1} \times \vec{n_2} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & -1 & 1 \\ 1 & 1 & 0 \end{vmatrix} = -\vec{i} + \vec{j} + 2\vec{k}$$

$\Rightarrow D: \frac{x-2}{-1} = \frac{y+3}{1} = \frac{z-4}{2} = t$

$\rightarrow$ direcția lui d $\Rightarrow$ direcția lui D
$d \parallel D$

L5) $M(2,1,-3)$ $\quad P: x-3y-7z+5=0$



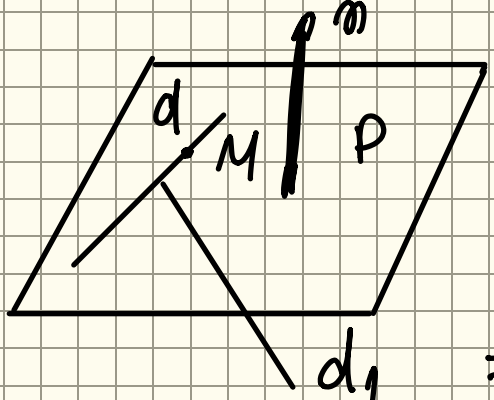
$\vec{n} = \vec{i} - 3\vec{j} - 7\vec{k}$

$\frac{x-2}{1} = \frac{y-1}{-3} = \frac{z+3}{-7} = t$

L6) $P: x - y + 3z - 5 = 0$

$M(1,-1,1)$

$d_1: \begin{cases} x = 3 + t \\ y = 2 - t \\ z = -5 \end{cases}, t \in \mathbb{R}$



$\vec{n} = \vec{i} - \vec{j} + 3\vec{k}$

$\vec{n}_{d_1} = \vec{i} - \vec{j}$

$\Rightarrow \vec{n} \times \vec{n}_{d_1} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & -1 & 3 \\ 1 & -1 & 0 \end{vmatrix}$

$= 3\vec{i} - 3\vec{j} \Rightarrow d: \frac{x-1}{3} = \frac{y+1}{-3} = \frac{z-1}{0} = t$

$\Rightarrow d: \begin{cases} x = 1 + 3t \\ y = -1 - 3t \\ z = 1 \end{cases}$